SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name		Venkataramana Veeramsetty	
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Course Code	24CS002PC215	**Course Title**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	Week7 - Thursday	**Time(s)**	
Duration	2 Hours	**Applicable to Batches**	
AssignmentNumber:13.1(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 13: Code Refactoring – Improving Legacy Code with AI Suggestions** **Lab Objectives:** • Identify code smells and inefficiencies in legacy Python scripts. • Use AI-assisted coding tools to **refactor** for readability,	Week7 - Thursday

maintainability, and performance.

- Apply **modern Python best practices** while ensuring output correctness.

Task 1

- **Task:** Refactor repeated loops into a cleaner, more Pythonic approach.

Instructions:

- Analyze the legacy code.
- Identify the part that uses loops to compute values.
- Refactor using **list comprehensions** or helper functions while keeping the output the same.

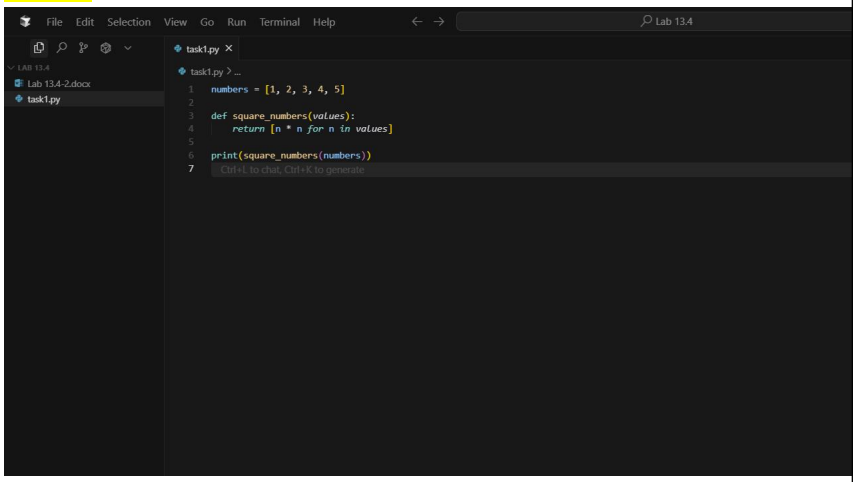
Legacy Code:

```
numbers = [1, 2, 3, 4, 5]
squares = []
for n in numbers:
    squares.append(n ** 2)
print(squares)
```

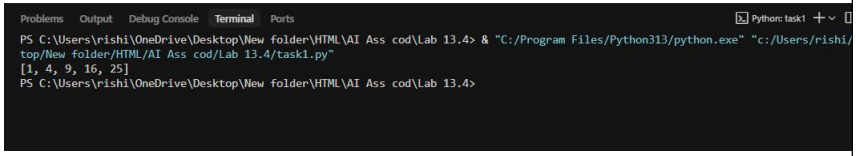
Expected Output:

```
[1, 4, 9, 16, 25]
```

CODE:



Output



Task 2

Task: Simplify string concatenation.

Instructions:

- Review the loop that builds a sentence using +=.
- Refactor using " ".join() to improve efficiency and readability.

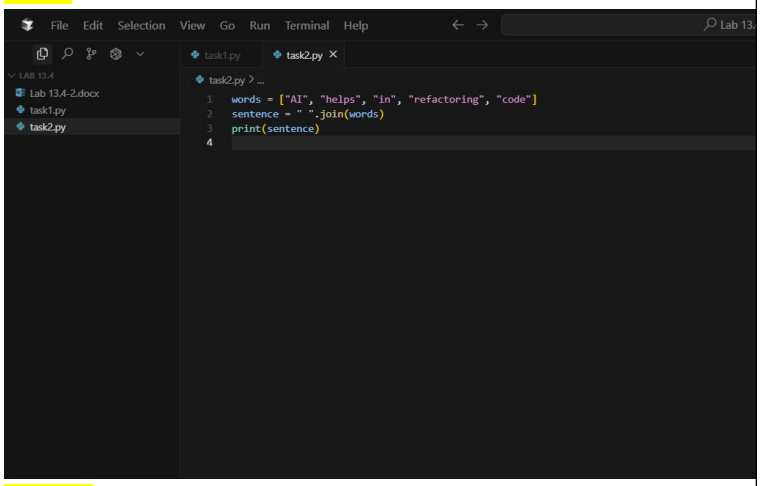
Legacy Code:

```
words = ["AI", "helps", "in", "refactoring", "code"]
sentence = ""
for word in words:
    sentence += word + " "
print(sentence.strip())
```

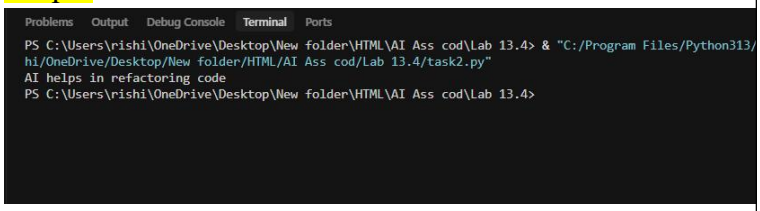
Expected Output:

AI helps in refactoring code

- Code



Output



Task 3

Task: Replace manual dictionary lookup with a safer method.

Instructions:

- Check how the code accesses dictionary keys.
- Use .get() or another Pythonic approach to handle missing keys gracefully.

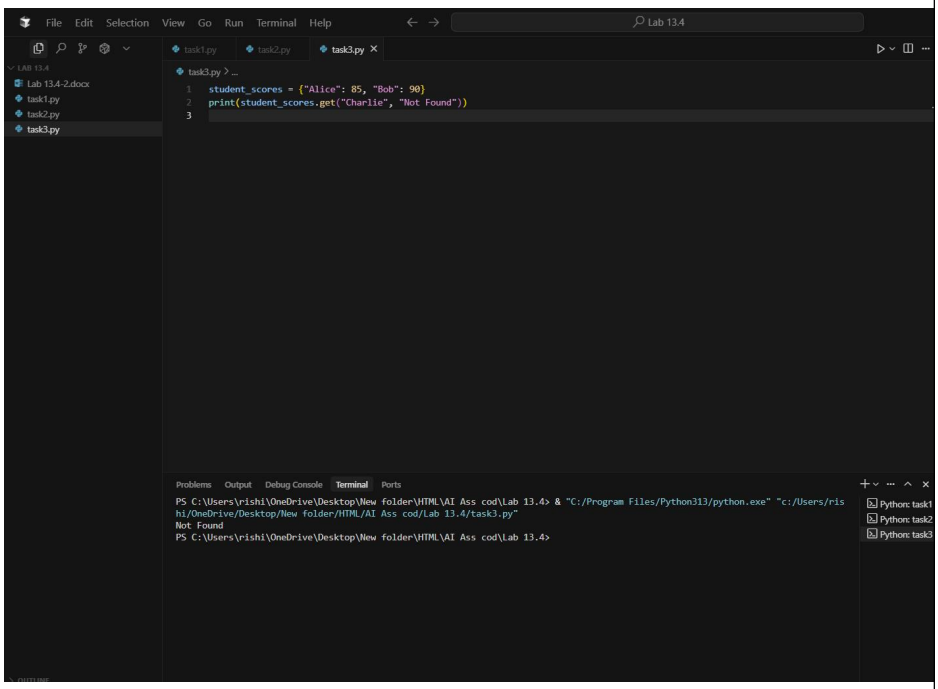
Legacy Code:

```
student_scores = {"Alice": 85, "Bob": 90}
if "Charlie" in student_scores:
    print(student_scores["Charlie"])
else:
    print("Not Found")
```

Expected Output:

Not Found

Code within the output



Task 4

Task: Refactor repetitive if-elif blocks.

Instructions:

- Examine multiple if-elif statements for operations.
- Refactor using **dictionary mapping** to make the code scalable and clean.

Legacy Code:

```
operation = "multiply"
a, b = 5, 3

if operation == "add":
    result = a + b
elif operation == "subtract":
```

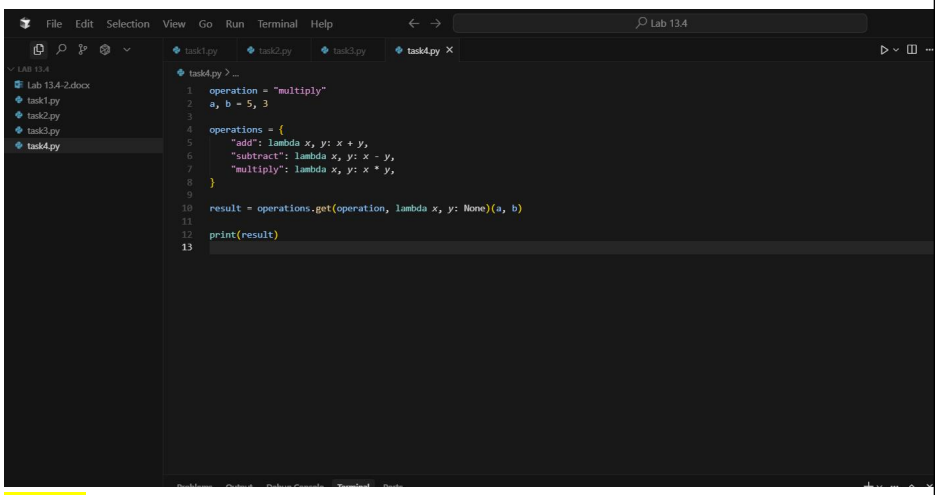
```
    result = a - b
elif operation == "multiply":
    result = a * b
else:
    result = None

print(result)
```

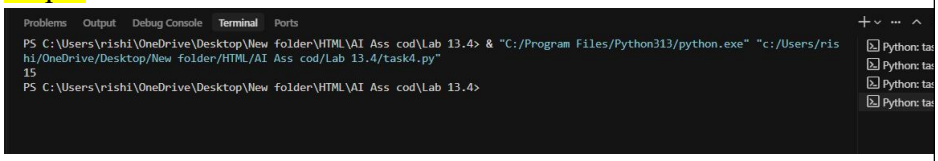
Expected Output:

15

code



output



Task 5

Task: Optimize nested loops for searching.

Instructions:

- Identify the nested loop used to find an element.
- Refactor using Python's in keyword or other efficient search techniques.

Legacy Code:

```
items = [10, 20, 30, 40, 50]
found = False
```

```
for i in items:
    if i == 30:
        found = True
        break
print("Found" if found else "Not Found")
```

Expected Output:

Found

Code and output: